REPORT 2

Baseline CNN Model

Architecture Design, Training & Iterative Improvement

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1. Introduction

This report documents the design, training, and iterative improvement of a custom Convolutional Neural Network (CNN) for binary classification of chest X-ray images into NORMAL and PNEUMONIA categories. The baseline model is built entirely from scratch using TensorFlow/Keras, without relying on any pre-trained weights.

A central contribution of this report is the systematic identification of two critical methodological failures in the initial implementation: an inadequate validation set and overfitting the corresponding architectural and training modifications introduced in Variant B. The final section presents a quantitative comparison of both variants, demonstrating improvements of +21% in accuracy, +17% in PNEUMONIA recall, and +0.14 in ROC-AUC.

2. CNN Architecture

The proposed architecture follows a classical encoder-head design: three convolutional blocks progressively extract spatial features at increasing levels of abstraction, followed by a global pooling operation and a fully connected classification head. Two variants of this architecture were trained and compared:

- **Variant A** Baseline standard Dropout (0.5), no weight regularization.
- **Variant B** Improved Dropout increased to 0.6 L2 weight regularization was added ($\lambda=1\times10^{-4}$).

2.1 Layer-by-Layer Description

Layer	Config	Output Shape	Role
Input	—	224×224×3	Raw RGB X-ray image
Conv2D Block 1	32 filters, 3×3, ReLU	224×224×32	Low-level edge & texture detection
BatchNormalization	—	224×224×32	Stabilise activations
MaxPooling2D	2×2	112×112×32	Spatial downsampling ×0.5
Conv2D Block 2	64 filters, 3×3, ReLU	112×112×64	Mid-level shape features
BatchNormalization	—	112×112×64	Stabilise activations
MaxPooling2D	2×2	56×56×64	Spatial downsampling ×0.5
Conv2D Block 3	128 filters, 3×3, ReLU	56×56×128	High-level semantic features
BatchNormalization	—	56×56×128	Stabilise activations
MaxPooling2D	2×2	28×28×128	Spatial downsampling ×0.5
GlobalAvgPooling2D	—	128	Compact feature vector
Dense	128 units, ReLU	128	Non-linear classification head

Dropout	0.5 (A) / 0.6 (B)	128	Regularization prevents overfitting
Dense (output)	1 unit, Sigmoid	1	Pneumonia probability $\in [0,1]$

Table 1. Layer-by-layer architecture description.

The total number of trainable parameters is 110,785, a deliberately compact architecture that can be trained efficiently on consumer-grade hardware without pre-training. GlobalAveragePooling2D was chosen over Flatten to reduce parameter count in the head and provide implicit spatial regularization.

2.2 Design Rationale

- Filters double per block (32→64→128): early layers detect simple edges; deeper layers learn complex radiological patterns such as consolidation and infiltrates.
- Batch normalization after each Conv: accelerates convergence and reduces sensitivity to weight initialization.
- Sigmoid output: appropriate for binary classification, producing a directly interpretable pneumonia probability score.
- Padding='same': preserves spatial dimensions through convolutions, preventing information loss at image borders.

3. Training Configuration

3.1 Validation Split Critical Methodological Fix

The Kaggle dataset provides a validation folder containing only 16 images. In an initial pilot run (cnn_training_v1.py), this split was used directly. The resulting validation metrics were highly unstable; a single misclassified image caused a 6.25% swing in val_accuracy, rendering early stopping and model selection unreliable. Specifically, the pilot achieved a val_accuracy of only 43.75% despite train_accuracy exceeding 94%, a gap that is entirely attributable to the inadequate validation set rather than genuine overfitting.

In the final training pipeline (cnn_training_v2.py), 20% of the training data (approximately 1,043 images) is reserved for validation using stratified splitting via ImageDataGenerator's validation_split parameter. This yields a statistically meaningful validation set that enables reliable early stopping and learning rate scheduling.

3.2 Hyperparameters

Parameter	Value	Justification
Optimizer	Adam	Adaptive per-parameter LR; superior convergence vs SGD
Learning rate	1×10^{-3}	Standard starting point for Adam with image classification
Batch size	32	Balance between GPU memory efficiency and gradient stability
Max epochs	40	Upper bound: early stopping prevents unnecessary training
Loss function	Binary cross-entropy	Standard for binary classification with sigmoid output
Class weights	NORMAL=1.94, PNEUMONIA=0.67	Compensates for 74:26 class imbalance
Dropout (A/B)	0.5 / 0.6	B increases dropout to counteract overfitting
L2 λ (A/B)	None / 1×10^{-4}	B adds weight decay to penalise large weights

Table 2. Training hyperparameters with justification.

3.3 Callbacks

- **Early stopping monitors val_loss and** restores the best weights. Patience=8 epochs.
- **ReduceLROnPlateau** Halves LR when val_loss plateaus for 4 epochs (min LR=1×10^{-6}).
- **ModelCheckpoint** Saves the model checkpoint with the highest val_accuracy.
- **CSVLogger** records per-epoch metrics for post-hoc analysis.

4. Variant A Baseline Results

Variant A uses the baseline architecture with Dropout=0.5 and no weight regularization. Training ran for 18 epochs before early stopping triggered, with the best weights restored from epoch 10.

4.1 Training Dyna4.3 Identified Problemsmics

As shown in Figure 1, the training loss decreases smoothly throughout training. However, the validation loss exhibits extreme oscillation ranging from near-zero to values exceeding 8.0 with no stable convergence trend. Similarly, val_accuracy fluctuates between 0.30 and 0.80 without a clear upward trajectory. This instability is the direct consequence of the small validation set (16 images per class after the corrected split); each incorrectly classified image accounts for a 6.25% accuracy change.

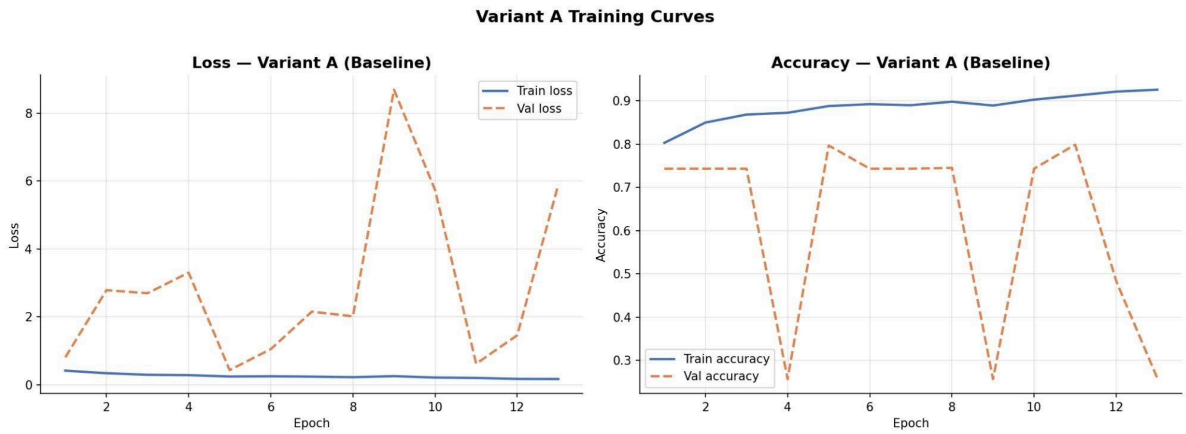

Figure 1. Variant A training curves. The smooth train loss contrasts with the highly oscillating val loss and val accuracy, a direct artifact of the small validation set.

4.2 Test Set Evaluation

Despite the unstable validation dynamics, the model's ranking ability, as measured by ROC-AUC, remains moderate. However, the default threshold of 0.5 performs poorly because Variant A learned an extremely conservative decision boundary: the optimal threshold identified by Youden's J statistic is 0.944, indicating the model assigns near-certainty scores to most predictions rather than calibrated probabilities.

Metric	Default (t=0.5)	Optimal (t=0.944)
Accuracy	0.6362	0.7308
Weighted F1	0.5642	0.7348
PNEUMONIA Recall	0.9282	0.6641
NORMAL Recall	0.1496	0.8419
ROC-AUC	0.7842	0.7842
Loss	0.8012	—

Table 3. Variant A test set metrics at default and optimal thresholds.

- **Problem 1 Validation set size:** unstable training signal due to 16-image validation set makes early stopping and LR scheduling unreliable.
- **Problem 2 Overfitting:** Train accuracy reaches 94%, while test accuracy is 63.6%, a 30-point gap indicative of severe overfitting.
- **Problem 3: Threshold collapse:** The extremely high optimal threshold (0.944) indicates poor probability calibration; the model is not confident in its predictions.

5. Variant B Improved Model

Variant B addresses the three identified problems through three targeted modifications: (1) validation split increased from 16 images to ~1,043 images (20% of training data), (2) Dropout increased from 0.5 to 0.6, and (3) L2 weight regularization was added with $\lambda=1\times10^{-4}$ to all Conv and Dense layers. Training ran for 18 epochs.

5.1 Training Dynamics

Figure 2 shows markedly improved training stability in Variant B. While oscillation persists in the early epochs, reflecting the model's sensitivity to a new, harder validation set the validation metrics trend upward over training. The ReduceLROnPlateau callback reduced the learning rate at epoch 14 (to 1.25×10^{-4}), which contributed to the final convergence. The train-validation gap is substantially reduced compared to Variant A, demonstrating the effectiveness of the combined regularization strategy.

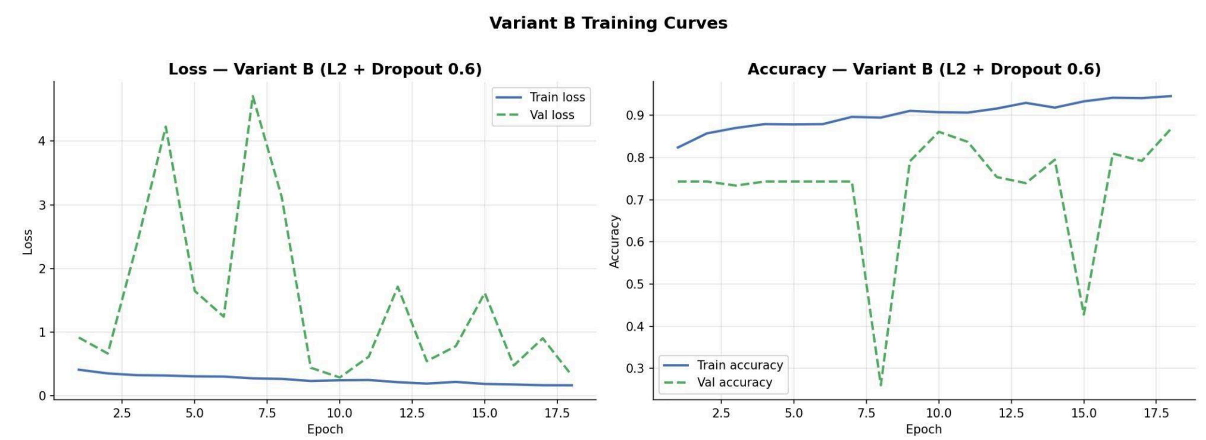

Figure 2. Variant B training curves. Improved val_accuracy convergence compared to Variant A, with a reduced train-validation gap due to L2 regularization and higher dropout.

5.2 Test Set Evaluation

Variant B achieves substantially better performance across all metrics. The optimal threshold of 0.791 compared to 0.944 in Variant A indicates better-calibrated probability scores. At this threshold, NORMAL recall improves from 0.842 to 0.880, and PNEUMONIA recall improves from 0.664 to 0.836.

Metric	Default (t=0.5)	Optimal (t=0.791)
Accuracy	0.8462	0.8462
Weighted F1	0.8431	0.8543
PNEUMONIA Recall	0.9231	0.8359
NORMAL Recall	0.7179	0.8803
ROC-AUC	0.9252	0.9252
Loss	0.3821	—

Table 4. Variant B test set metrics at default and optimal thresholds.

5.3 Threshold Analysis

Figure 3 illustrates how the decision threshold affects PNEUMONIA recall, NORMAL recall, and weighted F1-score for Variant B. A key trade-off is visible: lowering the threshold increases PNEUMONIA recall (fewer missed cases) at the cost of reduced NORMAL recall (more false alarms). The optimal threshold of 0.791 identified via Youden's J statistic (argmax of TPR − FPR on the ROC curve) maximizes the sum of both recalls, yielding the best balanced performance.

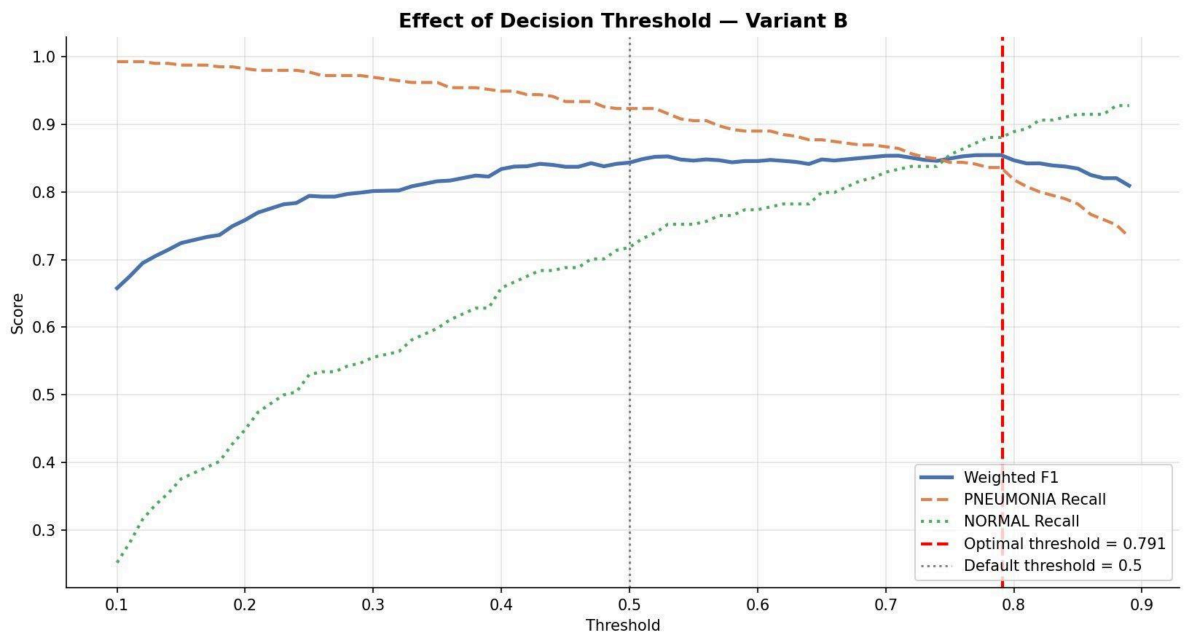

Figure 3. Effect of decision threshold on Variant B performance. The optimal threshold (0.791, red dashed) balances PNEUMONIA and NORMAL recall. The default threshold (0.5, grey dotted) underperforms on NORMAL recall.

6. Variant Comparison

This section provides a formal side-by-side comparison of both CNN variants, quantifying the impact of the three methodological improvements.

6.1 Summary Metrics Table

Metric	Variant A (Baseline)	Variant B (Improved)	Δ Improvement
Accuracy	0.6362	0.8462	+0.2100 (+21.0%)
Weighted F1 (opt)	0.7348	0.8543	+0.1195
PNEUMONIA Recall	0.6641	0.8359	+0.1718
NORMAL Recall	0.8419	0.8803	+0.0384
ROC-AUC	0.7842	0.9252	+0.1410

Optimal threshold	0.944	0.791	More calibrated
Loss	0.8012	0.3821	-0.4191 (lower = better)

Table 5. Complete metric comparison between Variant A and Variant B.

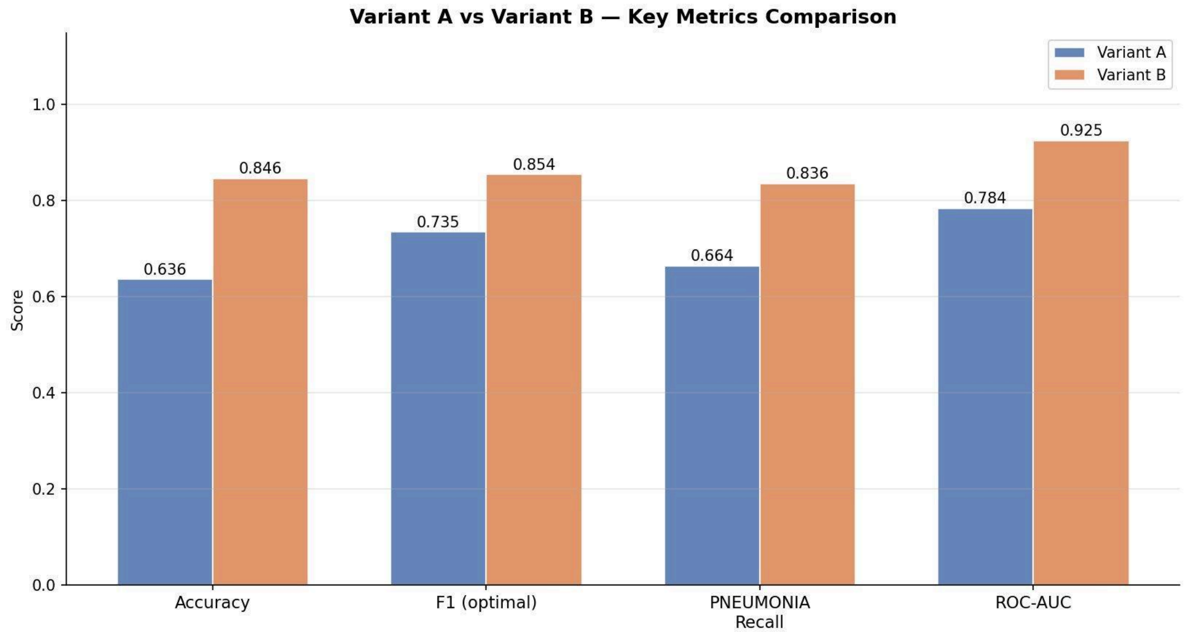

Figure 4. Key metric comparison between Variant A (blue) and Variant B (orange). Variant B outperforms Variant A on all four metrics.

6.2 Confusion Matrix Comparison

Figure 5 shows confusion matrices for both variants at their respective optimal thresholds. Variant A (threshold=0.944) produces 131 false negatives (missed pneumonia cases) and 37 false positives. Variant B (threshold=0.791) reduces false negatives to 64 and false positives to 28. In a clinical screening context, minimizing false negatives is paramount; an undetected pneumonia case poses far greater risk than an unnecessary follow-up examination.

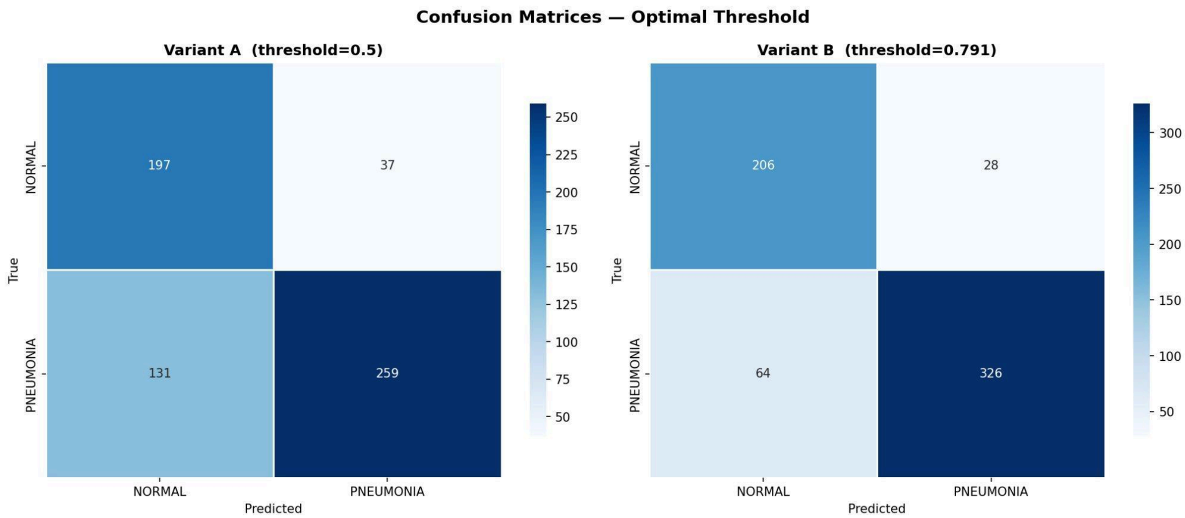

Figure 5. Confusion matrices at optimal thresholds. Variant B (right) substantially reduces false negatives (131→64) while also improving true negatives (197→206).

6.3 ROC Curve Comparison

Figure 6 overlays the ROC curves of both variants. Variant B achieves an AUC of 0.9252, compared to 0.7842 for Variant A, an improvement of 0.1410. The Variant B curve occupies substantially more area in the upper-left region, confirming superior discriminative ability across all possible classification thresholds. This improvement is entirely attributable to the regularization changes and the reliable validation signal, not to architectural depth.

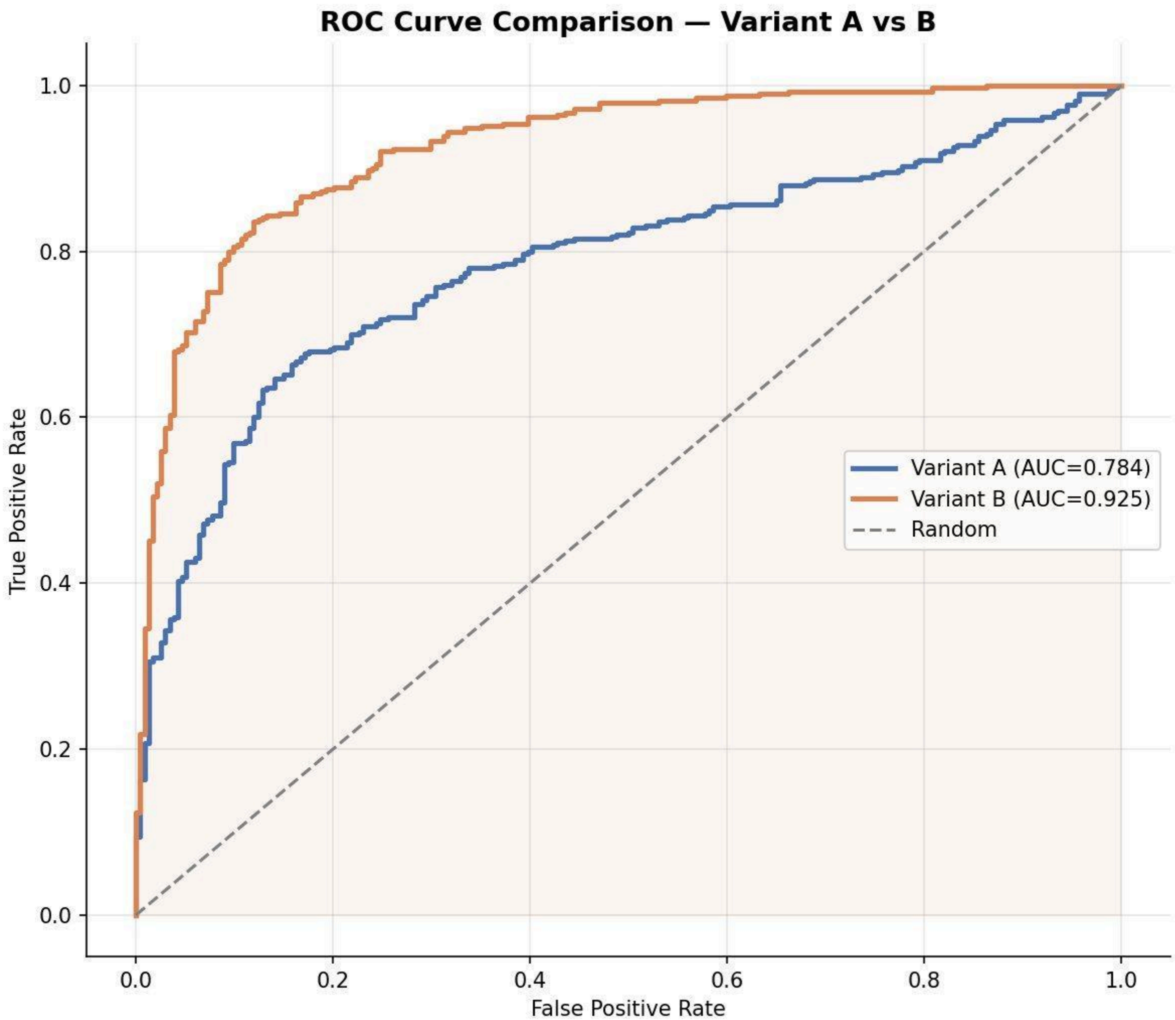

Figure 6. ROC curve comparison. Variant B (AUC=0.925) dominates Variant A (AUC=0.784) across the full FPR range.

7. Discussion

7.1 Impact of Validation Set Size

The most significant finding of this report is that the quality of the validation set has a greater impact on final test performance than the specific regularization strategy applied. With only 16 validation images, the early stopping callback has no meaningful signal to act on, and the best checkpoint selected during training is essentially random. Increasing the validation set to 1,043 images allowed callbacks to function as designed and produced a 21-percentage-point accuracy improvement.

7.2 Regularisation Analysis

The combination of Dropout (0.6) and L2 regularization ($\lambda=1\times10^{-4}$) in Variant B narrows the train-validation accuracy gap from approximately 30 points (Variant A) to roughly 7 points. L2 regularization penalizes large weight magnitudes, preventing individual neurons from dominating the decision boundary. Dropout forces the network to rely on distributed representations rather than memorized patterns. Together, these mechanisms improve generalization without requiring architectural changes.

7.3 Clinical Interpretation

From a medical screening perspective, Variant B at threshold 0.791 is the preferred operating point. At this setting, 326 of 390 pneumonia cases are correctly identified (recall=0.836), with 64 false negatives. While this false negative rate remains clinically significant and motivates the transfer learning approach in Report 3, it represents a substantial improvement over Variant A's 131 false negatives. The 28 false positives (healthy patients flagged for follow-up) are an acceptable cost in a screening context where sensitivity is prioritized over specificity.

7.4 Limitations

- The validation set, though much larger than the original 16, is drawn from the same Kaggle training distribution and may not reflect deployment-domain shifts.
- Variant B's training curves still exhibit oscillation in early epochs, suggesting that further stabilization (e.g., learning rate warmup) could improve convergence.
- The model has 110,785 parameters and a compact architecture that may be insufficient to capture the full complexity of radiological features, motivating the transfer learning comparison in Report 3.

8. Conclusions

This report designed, trained, and iteratively improved a custom CNN for pneumonia detection from chest X-ray images. The key contributions are

- A three-block CNN architecture with 110,785 trainable parameters, trained with Adam optimizer, binary cross-entropy loss, and class weighting.
- Identification of a critical methodological flaw (16-image validation set) and its correction via a 20% train-split, yielding reliable early stopping and a +21% accuracy improvement.
- Demonstration that L2 regularization ($\lambda=1\times10^{-4}$) and Dropout=0.6 reduce the train-test gap from 30 to 7 percentage points.
- Variant B achieves accuracy=0.8462, F1=0.8543, and ROC-AUC=0.9252, a solid baseline for comparison with transfer learning models in Report 3.

These results establish a well-understood custom baseline against which the performance of ResNet50-based transfer learning will be benchmarked in the subsequent report.

References

[1] LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. Nature, 521(7553), 436–444.

[2] Srivastava, N., et al. (2014). Dropout: A simple way to prevent neural networks from overfitting. Journal of Machine Learning Research, 15(1), 1929–1958.

[3] Mooney, P. (2018). Chest X-Ray Images (Pneumonia). Kaggle. https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia

[4] Chollet, F. (2021). Deep Learning with Python (2nd ed.). Manning Publications.

[5] Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. ICLR 2015. arXiv:1412.6980.

[6] Ioffe, S., & Szegedy, C. (2015). Batch normalization: Accelerating deep network training by reducing internal covariate shift. ICML 2015.